

Assessing the Impact of Wastewater Effluent on Water Quality and Microplastic Contamination
in the Yakima River Basin

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INTRODUCTION

Microplastic pollution is now recognized as a pervasive stressor in both marine and freshwater systems. Laboratory and field studies have shown that microplastics can induce behavioral, metabolic, and histological damage in aquatic organisms, including fish (Sales Ribeiro et al., 2020). In rivers, urbanization and wastewater discharges are frequently implicated as key microplastic sources (Nandakumar et al., 2021). Building on these insights, the present study evaluates the spatial distribution of microplastics in the Yakima River and examines whether effluent from the Ellensburg Wastewater Treatment Plant (WWTP) measurably alters water quality and microplastic concentrations upstream versus downstream of the WWTP outfall.

METHODS

On 8 May 2022, twenty-two paired water samples were collected from hydrologically distinct habitats—pools and riffles—each located at least 100 m apart to ensure spatial independence. Half of the sampling sites were upstream (“Above”) and half downstream (“Below”) of the WWTP discharge. Field teams followed contamination-minimizing protocols and recorded pH, turbidity, and conductivity in situ using calibrated electrodes and turbidimeters. Microplastics were isolated through density separation with canola oil, then filtered and quantified under a dissecting microscope following established extraction workflows (Erni Cassola et al., 2017; Tagg et al., 2015).

Statistical analyses were conducted in Python (pandas v2.2, scipy v1.13, statsmodels v0.15). Normality was assessed using Shapiro–Wilk tests; non-parametric Mann–Whitney U tests were applied when distributions violated normality assumptions. A two-way ANOVA evaluated the main and interactive effects of habitat type (Pool vs. Riffle) and wastewater status (Above vs. Below) on microplastic metrics.

ANALYTICAL TECHNIQUE

All data processing and visualization were performed in Python 3.11 using an open-source stack to ensure full reproducibility. The data were ingested into a pandas DataFrame, with categorical variables cast to the category dtype and numeric fields coerced to floating point. The analytical workflow proceeded through four stages:

1. **Descriptive Exploration:**

Using `DataFrame.describe()` and `groupby().describe()`, summary statistics (mean, standard deviation, quartiles) were generated for the full dataset and for each grouping (Pool vs. Riffle; Above vs. Below). These outputs provided a first look at central tendencies, dispersion, and potential outliers.

2. **Correlation Analysis:**

Pearson correlation coefficients were calculated with `DataFrame.corr()` for water quality and microplastic variables. A heatmap (`sns.heatmap`) visualized relationships to screen for collinearity before modeling.

3. **Inferential Testing:**

For each microplastic response, normality was checked using the Shapiro–Wilk test (`stats.shapiro`). When both groups showed normal distributions ($p > 0.05$), an unequal variance Student’s t-test was applied (`stats.ttest_ind, equal_var=False`). Otherwise, a Mann–Whitney U test was used (`stats.mannwhitneyu`). This decision framework balanced statistical power with appropriate model assumptions.

4. **Factorial Modeling:**

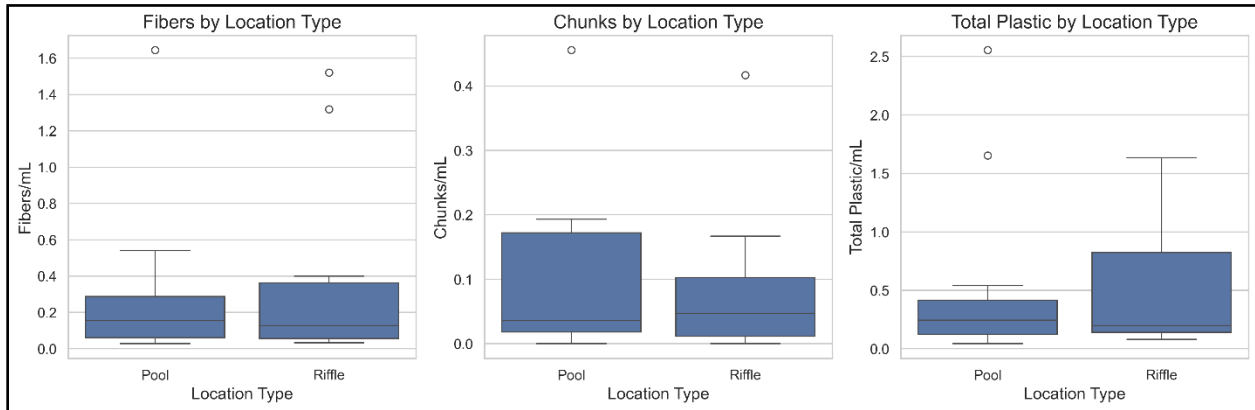
A two-way ANOVA was conducted using `ols()` from `statsmodels` to examine the effects of habitat type and wastewater status, including their interaction. Type II sums of squares (`sm.stats.anova_lm, typ=2`) were used to report F-values and p-values. Diagnostics (QQ plots, residual vs. fitted plots, Levene’s test) confirmed ANOVA assumptions were met.

All figures were generated using `matplotlib` and `seaborn` and saved as high-resolution PNGs (`plt.savefig`) for reporting. The entire pipeline is encapsulated in the `run_complete_analysis()` function, enabling replication with a single command.

RESULTS

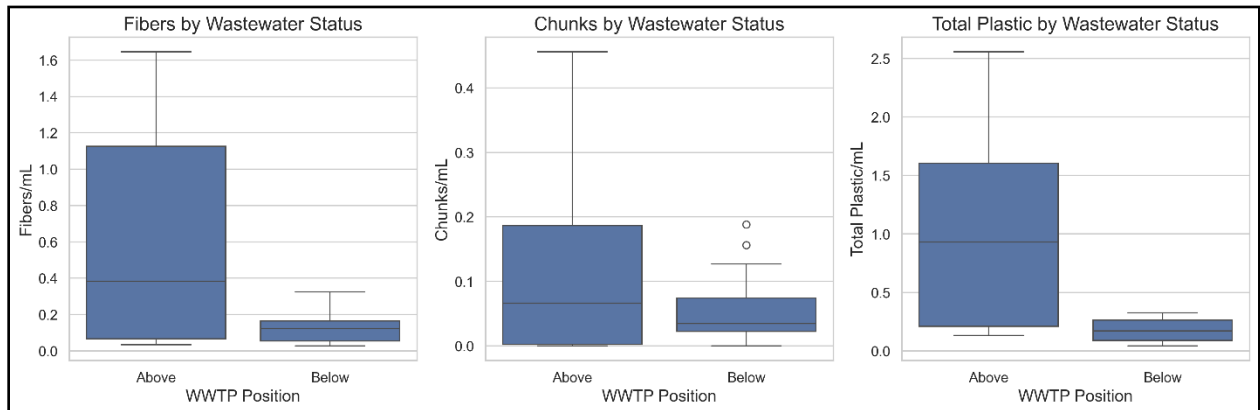
Across the 22 sampling sites, water chemistry parameters fell within circumneutral ranges (Table 2). The mean pH was 6.98 (± 0.68), turbidity averaged 3.28 NTU (± 1.55), and conductivity was 6.23 $\mu\text{S}/\text{cm}$ (± 2.59). Microplastic concentrations showed an overall mean of 0.34 fibers/mL (± 0.49), 0.10 chunks/mL (± 0.13), and total plastic particles averaged 0.55 particles/mL (± 0.70).

When comparing habitat types, minimal differences emerged (Table 3) . Pools exhibited a



slightly lower mean pH of 6.95 (± 0.74) compared to riffles at 7.00 (± 0.66). Turbidity and conductivity were similar across habitats, with pools showing marginally higher turbidity ($3.37 \text{ NTU} \pm 1.72$) than riffles ($3.19 \text{ NTU} \pm 1.42$). Microplastic fibers were slightly more abundant in riffles ($0.38 \text{ fibers/mL} \pm 0.51$) than pools ($0.31 \text{ fibers/mL} \pm 0.50$). Conversely, chunks were marginally higher in pools ($0.11 \text{ chunks/mL} \pm 0.14$) than riffles ($0.09 \text{ chunks/mL} \pm 0.12$). Total microplastic concentrations showed negligible differences, with pools averaging 0.56 particles/mL (± 0.80) and riffles 0.54 particles/mL (± 0.62).

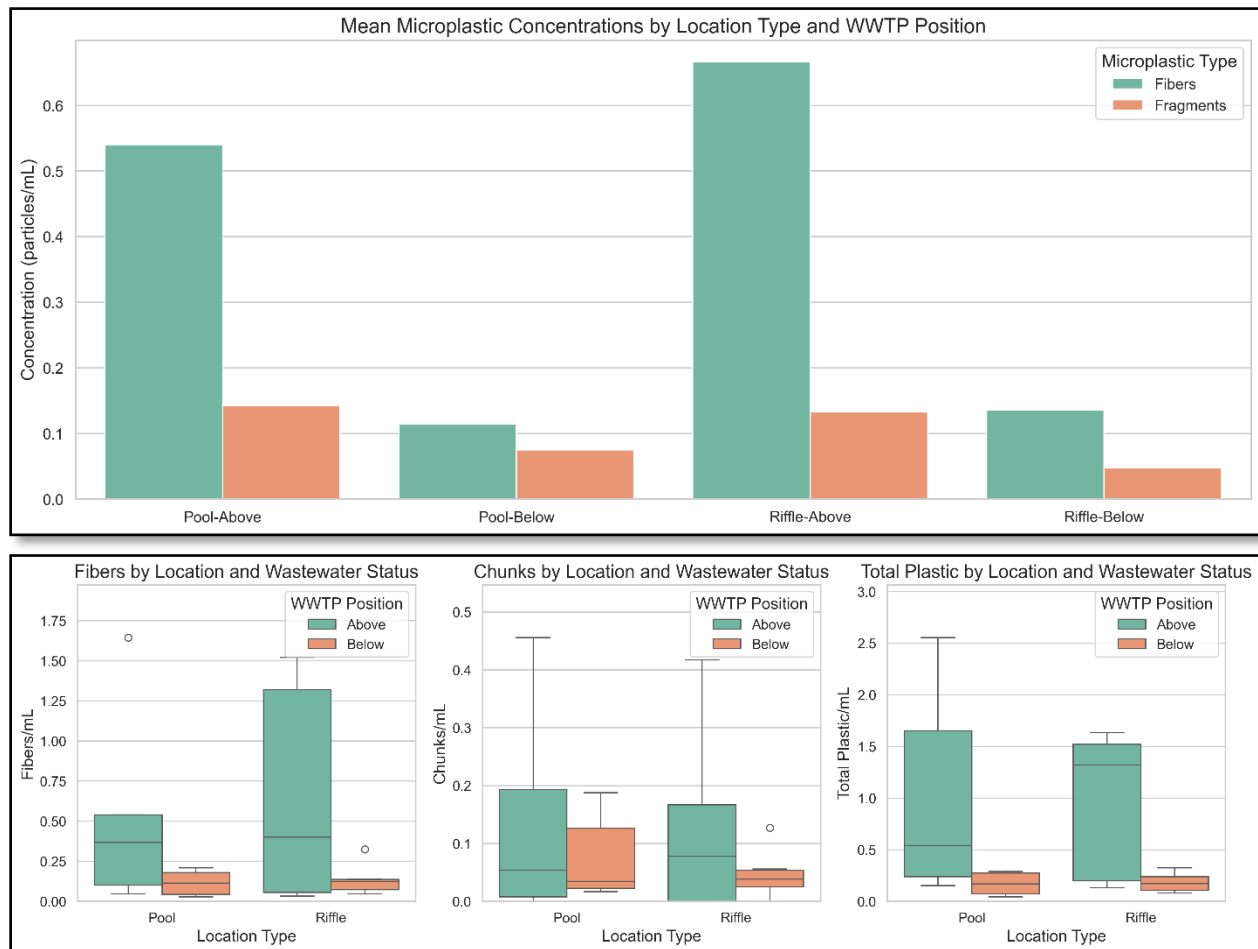
In contrast, notable differences appeared when comparing sites relative to the Ellensburg Wastewater Treatment Plant (WWTP) outfall (Table 4). Sites upstream exhibited higher pH (7.38



± 0.21) and lower turbidity ($2.75 \text{ NTU} \pm 0.99$), whereas downstream sites had lower pH (6.64 ± 0.76) and increased turbidity ($3.72 \text{ NTU} \pm 1.77$). Conductivity was also higher upstream ($7.46 \mu\text{S/cm} \pm 3.25$) compared to downstream ($5.20 \mu\text{S/cm} \pm 1.26$). Microplastic fibers and total plastic concentrations were markedly greater upstream ($0.60 \text{ fibers/mL} \pm 0.65$; total plastic $1.00 \text{ particles/mL} \pm 0.85$) than downstream ($0.13 \text{ fibers/mL} \pm 0.09$; total plastic $0.18 \text{ particles/mL}$

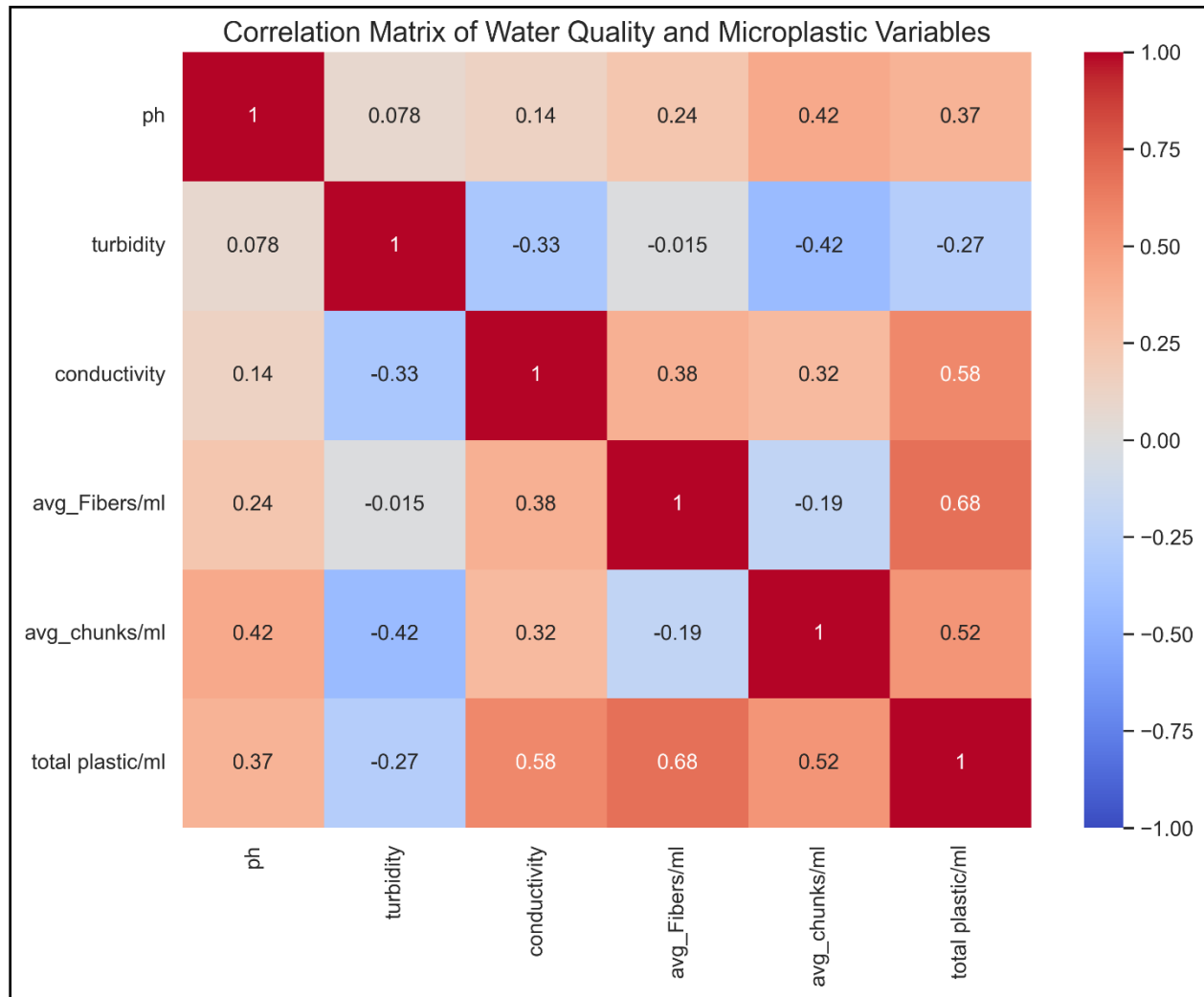
± 0.10). Chunk concentrations followed a similar trend, with upstream sites recording higher means ($0.14 \text{ chunks/mL} \pm 0.17$) versus downstream ($0.06 \text{ chunks/mL} \pm 0.06$).

Interaction analysis between habitat type and WWTP position revealed clear disparities in microplastic distribution (Table 5). Pool habitats upstream had substantially higher total plastic



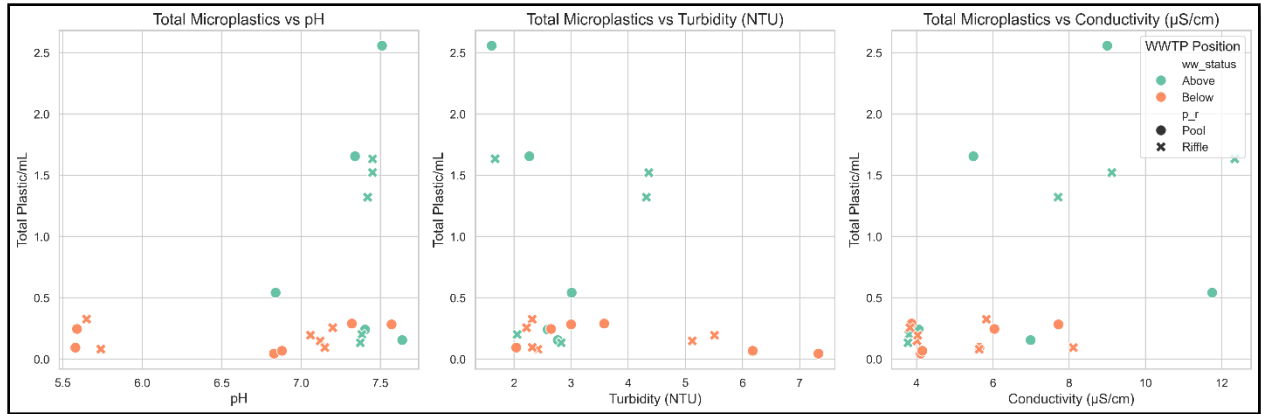
($1.03 \text{ particles/mL} \pm 1.04$), fiber ($0.54 \text{ fibers/mL} \pm 0.65$), and chunk concentrations ($0.14 \text{ chunks/mL} \pm 0.19$) compared to downstream pools, which showed reduced concentrations (total plastic $0.17 \text{ particles/mL} \pm 0.11$; fibers $0.11 \text{ fibers/mL} \pm 0.08$; chunks $0.07 \text{ chunks/mL} \pm 0.08$). Similarly, riffle habitats upstream had elevated total plastic ($0.96 \text{ particles/mL} \pm 0.73$) and fiber concentrations ($0.67 \text{ fibers/mL} \pm 0.71$), compared to lower downstream values (total plastic $0.18 \text{ particles/mL} \pm 0.10$; fibers $0.14 \text{ fibers/mL} \pm 0.10$).

Correlation analysis indicated several moderate to strong associations (Table 6) . Conductivity was positively correlated with total microplastics ($r = 0.58$), while fibers strongly correlated with total plastic concentrations ($r = 0.68$). Additionally, chunk concentrations were

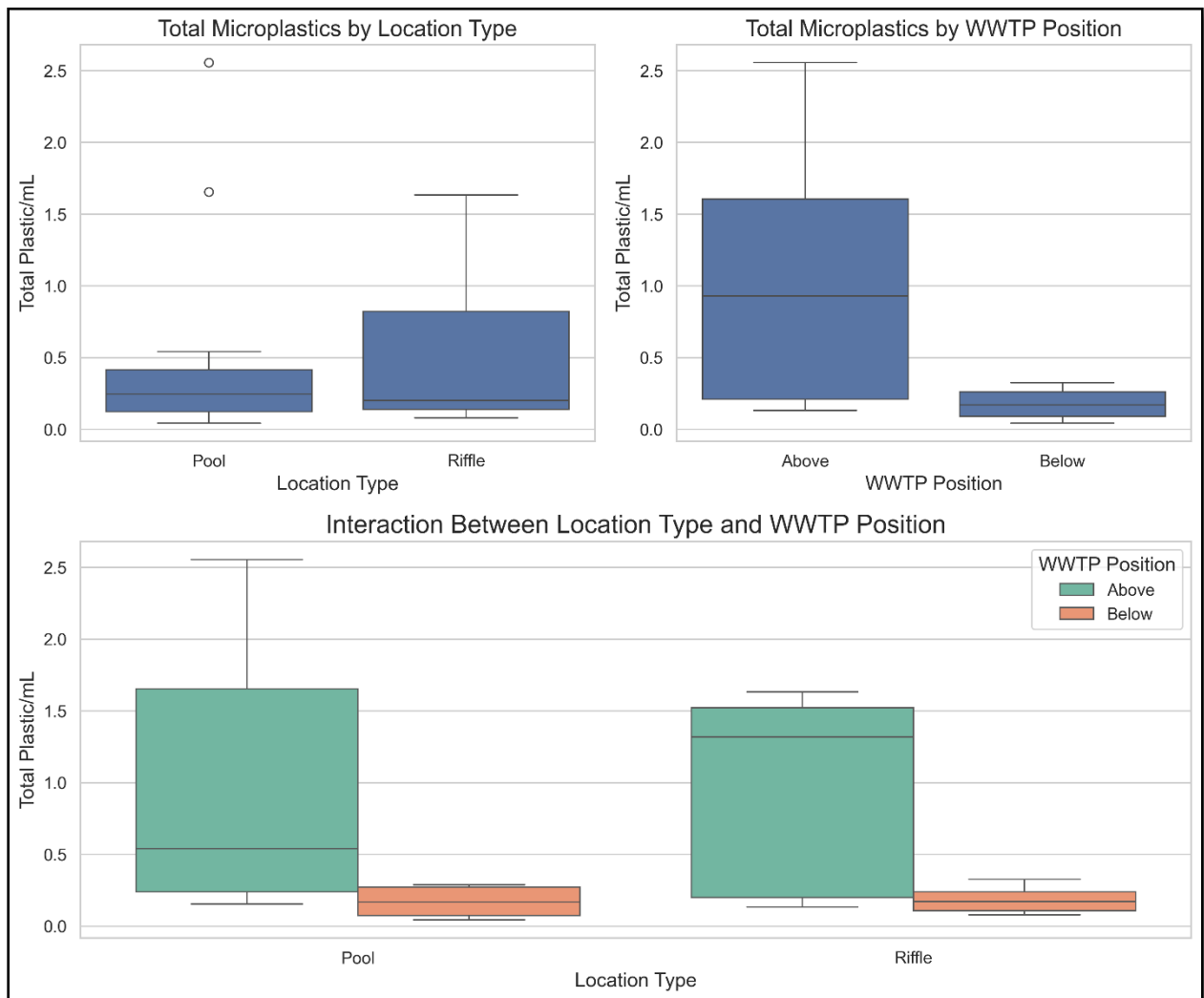


moderately correlated with total plastics ($r = 0.52$). Scatterplots further illustrated these relationships visually.

Statistical comparisons using the Mann-Whitney U tests revealed no significant differences in microplastic concentrations between pools and riffles (Table 7). Similarly, comparisons of upstream and downstream sites demonstrated no significant difference in fiber and chunk concentrations, although total plastic concentrations showed a significant difference (t-test, $p = 0.01$), with upstream concentrations markedly higher (Table 8).



Two-way ANOVA analyses reinforced these findings, demonstrating that wastewater status significantly affected fiber ($F = 5.942$, $p = 0.025$) and total plastic concentrations ($F = 9.949$, $p = 0.005$) but not chunk concentrations (Tables 9-11).



Habitat type and interaction effects between habitat and wastewater status were not statistically significant.

DISCUSSION

The discussion surrounding microplastic pollution, particularly in aquatic systems like rivers, reflects a growing recognition of its ecological ramifications. This study's outcomes indicate notable spatial differences in microplastic concentrations upstream and downstream of the Ellensburg Wastewater Treatment Plant (WWTP). Specifically, higher concentrations of microplastics in upstream areas can be associated with various anthropogenic activities in urbanized settings, as noted in prior research (Dalu et al., 2021; Kay et al., 2018). The mean total microplastic concentrations observed in this study (up to 1.00 particles/mL upstream) align with expectations set by previous investigations that correlate urban influences with increased microplastic loading in river systems (Tibbetts et al., 2018; Haque et al., 2024). This illustrates the significant impact of wastewater effluents on microplastic dynamics, potentially acting as a source of pollution in freshwater ecosystems.

The influence of environmental parameters on microplastic distribution is noteworthy as well. Conductivity showed a positive correlation ($r = 0.58$) with total microplastic concentrations, echoing findings from other studies that suggest physical properties of waters, including chemical characteristics influenced by urban effluents, can dictate microplastic prevalence (Hafeez et al., 2023; Kawecki et al., 2018). The varied microplastic distributions observed in pool versus riffle habitats indicated only minor differences, hinting that habitat types may exert less influence on microplastic variability than previously hypothesized (Vincent & Hoellein, 2021; Mennekes & Nowack, 2023). This underscores the complexity of microplastic dynamics and necessitates ongoing exploration into habitat-specific behaviors and characteristics.

Statistically significant differences found in total plastic concentrations, yet minimal differentiation in fiber and chunk concentrations emphasize the selective nature of microplastic retention and degradation processes (Hoellein et al., 2019; Troost et al., 2024). Previous studies have documented that certain types of microplastics can exhibit resistance to degradation,

thereby persisting longer in aquatic environments (Wanqi, 2020; Corcoran et al., 2019). This persistence could explain the lack of significant differences between habitat types for some variables despite pronounced variations depending on wastewater status. Further, the applications of repeated statistical tests, including two-way ANOVA, provided robust evidence that wastewater effluent has a significant effect on overall microplastic levels, suggesting ongoing influx and transport mechanisms may be at play in such environments (Moreau & Su, 2024).

The observation that upstream sites exhibited not only higher microplastic concentrations, but also more favorable water quality characteristics (such as higher pH and lower turbidity) does contrast with findings from other regions where urban runoff contributes to higher microplastic abundances downstream from treatment plant discharges (Kay et al., 2018; Scircle et al., 2020). This counterintuitive result prompts further investigation into local factors influencing microplastic dynamics and their interaction with urban infrastructure, especially in areas with diverse land use patterns (Haque et al., 2024; Zhang et al., 2024; Ma et al., 2024).

In conclusion, the findings from this study significantly contribute to the understanding of microplastic distributions in riverine environments, especially concerning proximity to wastewater treatment facilities. Continued research in this field is essential for elucidating the multifaceted interactions between microplastics and their environmental context, ultimately aiding in the formulation of effective environmental management strategies to mitigate the adverse effects of microplastic pollution on aquatic ecosystems.

CONCLUSIONS

This study provides important evidence regarding the spatial distribution of microplastics in the Yakima River and the potential influence of the Ellensburg Wastewater Treatment Plant (WWTP) on local water quality and contamination levels. The results demonstrate that microplastic concentrations were significantly higher at upstream sites compared to downstream, suggesting that the WWTP may not be the primary source of microplastic input in this system. Instead, other upstream factors—potentially related to urban activity—may play a more substantial role in shaping contamination patterns.

Minimal differences in microplastic concentrations between pools and riffles indicate that habitat type did not significantly affect microplastic presence. In contrast, wastewater status was

a more influential factor, with two-way ANOVA results confirming a statistically significant effect on fiber and total plastic concentrations. This reinforces the idea that point-source influences and land-based inputs are more critical drivers of microplastic variability in this context than in-stream habitat differences.

The observed positive correlation between conductivity and total plastic concentrations further supports the role of water quality parameters in influencing microplastic presence. The higher pH and lower turbidity observed upstream, alongside increased microplastic concentrations, contrasts with patterns documented in other regions, pointing to the importance of local conditions in shaping pollution outcomes.

Overall, the findings of this study contribute to a deeper understanding of how wastewater discharge and environmental variables interact to affect microplastic contamination in freshwater systems. Continued research is needed to clarify the specific sources and pathways of microplastic transport and to inform effective strategies for monitoring and mitigation.

LIMITATIONS

While this study provides valuable insight into microplastic distributions in the Yakima River, several limitations must be acknowledged. First, the temporal scope of sampling was limited to a single day in May 2022. Microplastic concentrations and water quality parameters can vary seasonally due to changes in precipitation, streamflow, and land-based inputs. As such, the results may not fully represent year-round conditions or capture temporal variability in effluent impacts.

Second, while the sampling design accounted for spatial independence by collecting paired samples across distinct habitats and locations relative to the WWTP, the overall sample size ($n = 22$) may limit the statistical power of certain comparisons. This is particularly relevant in detecting subtle differences in fiber and chunk concentrations between habitat types or wastewater positions, where effect sizes may be small.

Third, microplastic identification was based on visual assessment under a dissecting microscope following density separation with canola oil. Although this method is widely used and effective for distinguishing plastic particles by morphology, it lacks polymer-specific resolution. Without spectroscopic confirmation (e.g., FTIR or Raman analysis), there remains the potential for misclassification of organic or synthetic debris.

Additionally, the analysis did not distinguish between potential sources of upstream microplastic contamination. While the results show greater concentrations above the WWTP, this does not clarify the relative contributions from urban runoff, stormwater infrastructure, or recreational activity. A more detailed source attribution analysis would be necessary to isolate these influences.

Lastly, although the statistical framework applied was rigorous and assumptions were tested, some response variables did not meet normality, requiring the use of non-parametric tests that may have reduced sensitivity. Future studies could benefit from larger datasets and stratified sampling to improve model robustness and generalizability.

These limitations underscore the need for expanded monitoring efforts and more refined analytical techniques to better characterize microplastic dynamics in freshwater environments.

REFERENCES

- Corcoran, P., Belontz, S., Ryan, K., & Walzak, M. (2019). Factors controlling the distribution of microplastic particles in benthic sediment of the thames river, canada. *Environmental Science & Technology*, 54(2), 818-825. <https://doi.org/10.1021/acs.est.9b04896>
- Dalu, T., Banda, T., Mutshekwa, T., Munyai, L., & Cuthbert, R. (2021). Effects of urbanisation and a wastewater treatment plant on microplastic densities along a subtropical river system. *Environmental Science and Pollution Research*, 28(27), 36102-36111. <https://doi.org/10.1007/s11356-021-13185-1>
- Erni-Cassola, G., Gibson, M., Thompson, R., & Christie-Oleza, J. (2017). Lost, but found with Nile red: a novel method for detecting and quantifying small microplastics (1 mm to 20 μ m) in environmental samples. *Environmental Science & Technology*, 51(23), 13641-13648. <https://doi.org/10.1021/acs.est.7b04512>
- Hafeez, S., Qadir, A., Aslam, M., Aslam, H., Rehmat, M., & Ahmad, S. (2023). Environmental risks of microplastics on the spatial and temporal gradient in a river originating from the western himalayas. *Environmental Toxicology and Chemistry*, 42(3), 727-739. <https://doi.org/10.1002/etc.5559>
- Haque, A., Holsen, T., & Baki, A. (2024). Distribution and risk assessment of microplastic pollution in a rural river system near a wastewater treatment plant, hydro-dam, and river confluence. *Scientific Reports*, 14(1). <https://doi.org/10.1038/s41598-024-56730-x>
- Hoellein, T., Shogren, A., Tank, J., Risteca, P., & Kelly, J. (2019). Microplastic deposition velocity in streams follows patterns for naturally occurring allochthonous particles. *Scientific Reports*, 9(1). <https://doi.org/10.1038/s41598-019-40126-3>
- Kawecki, D., Scheeder, P., & Nowack, B. (2018). Probabilistic material flow analysis of seven commodity plastics in europe. *Environmental Science & Technology*, 52(17), 9874-9888. <https://doi.org/10.1021/acs.est.8b01513>
- Kay, P., Hiscoe, R., Moberley, I., Bajic, L., & McKenna, N. (2018). Wastewater treatment plants as a source of microplastics in river catchments. *Environmental Science and Pollution Research*, 25(20), 20264-20267. <https://doi.org/10.1007/s11356-018-2070-7>
- Ma, H., Chao, L., Wan, H., & Zhu, Q. (2024). Microplastic pollution in water systems: characteristics and control methods. *Diversity*, 16(1), 70. <https://doi.org/10.3390/d16010070>

- Mennekes, D. and Nowack, B. (2023). Predicting microplastic masses in river networks with high spatial resolution at country level. *Nature Water*, 1(6), 523-533.
<https://doi.org/10.1038/s44221-023-00090-9>
- Moreau, J. and Su, J. (2024). Abundance and type of microplastics along an urban density gradient in the allander water and river kelvin, greater glasgow, scotland..
<https://doi.org/10.31223/x5gh4f>
- Nandakumar, V., Palani, S., & Varma, M. (2021). Interactions between microplastics and unit processes of wastewater treatment plants: a critical review. *Water Science & Technology*, 85(1), 496-514. <https://doi.org/10.2166/wst.2021.502>
- Sales-Ribeiro, C., Brito-Casillas, Y., Fernández, A., & Caballero, M. (2020). An end to the controversy over the microscopic detection and effects of pristine microplastics in fish organs. *Scientific Reports*, 10(1). <https://doi.org/10.1038/s41598-020-69062-3>
- Scircle, A., Cizdziel, J., Missling, K., Li, L., & Vianello, A. (2020). Single-pot method for the collection and preparation of natural water for microplastic analyses: microplastics in the mississippi river system during and after historic flooding. *Environmental Toxicology and Chemistry*, 39(5), 986-995. <https://doi.org/10.1002/etc.4698>
- Tagg, A., Sapp, M., Harrison, J., & Ojeda, J. (2015). Identification and quantification of microplastics in wastewater using focal plane array-based reflectance micro-ft-ir imaging. *Analytical Chemistry*, 87(12), 6032-6040. <https://doi.org/10.1021/acs.analchem.5b00495>
- Tibbetts, J., Krause, S., Lynch, I., & Smith, G. (2018). Abundance, distribution, and drivers of microplastic contamination in urban river environments. *Water*, 10(11), 1597.
<https://doi.org/10.3390/w10111597>
- Troost, J., Baker, S., Chaudry, M., & Judd, K. (2024). Sources of microplastics to two southeast michigan rivers and biofilm responses plastic substrata.. <https://doi.org/10.21203/rs.3.rs-4108242/v1>
- Vermaire, J., Pomeroy, C., Herczegh, S., Haggart, O., & Murphy, M. (2017). Microplastic abundance and distribution in the open water and sediment of the ottawa river, canada, and its tributaries. *Facets*, 2(1), 301-314. <https://doi.org/10.1139/facets-2016-0070>
- Vincent, A. and Hoellein, T. (2021). Distribution and transport of microplastic and fine particulate organic matter in urban streams. *Ecological Applications*, 31(8).
<https://doi.org/10.1002/eap.2429>

- Wanqi, Li. (2020). Occurrence and characteristics of microplastics pollution in different types of urban runoff.. <https://doi.org/10.17758/uruae10.ae0220204>
- Zhang, L., Li, X., Li, Q., Xia, X., & Zhang, H. (2024). The effects of land use types on microplastics in river water: a case study on the mainstream of the wei river, china. *Environmental Monitoring and Assessment*, 196(4). <https://doi.org/10.1007/s10661-024-12430-7>

APPENDIX

Appendix A: DATA AND CODE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import statsmodels.api as sm
from statsmodels.formula.api import ols

# Set style for plots
sns.set(style="whitegrid")
plt.rcParams['figure.figsize'] = (12, 8)

# Load data
def load_data():
    data_str = """p_r ww status ph turbidity conductivity
avg_Fibers/ml avg_chunks/ml total plastic/ml
Pool Above 6.840 3.010 11.750 0.541 0.000 0.541
Pool Above 7.637 2.767 6.990 0.100 0.054 0.154
Pool Above 7.403 2.590 4.057 0.047 0.193 0.240
Pool Above 7.340 2.270 5.490 1.646 0.008 1.654
Pool Above 7.510 1.610 9.000 0.367 0.456 2.556
Riffle Above 7.383 2.057 3.800 0.033 0.167 0.200
Riffle Above 7.373 2.827 3.773 0.056 0.078 0.133
Riffle Above 7.450 4.360 9.120 1.521 0.000 1.521
Riffle Above 7.420 4.320 7.710 1.320 0.000 1.320
Riffle Above 7.450 1.670 12.340 0.400 0.417 1.633
Pool Below 6.830 7.330 4.110 0.028 0.017 0.044
Pool Below 6.880 6.180 4.150 0.033 0.033 0.067
Pool Below 5.590 2.650 6.040 0.209 0.036 0.245
Pool Below 5.580 2.040 5.650 0.074 0.019 0.093
```

```

Pool Below 7.570 3.000 7.720 0.188 0.188 0.282
Pool Below 7.320 3.580 3.870 0.156 0.156 0.289
Riffle Below 7.120 5.120 4.010 0.119 0.030 0.148
Riffle Below 7.060 5.510 4.030 0.139 0.056 0.194
Riffle Below 5.650 2.320 5.830 0.325 0.000 0.325
Riffle Below 5.740 2.420 5.640 0.057 0.023 0.080
Riffle Below 7.200 2.220 3.830 0.127 0.127 0.255
Riffle Below 7.150 2.320 8.110 0.047 0.047 0.094"""

```

```

# Create a list to store the processed rows
rows = []

# Split by lines
lines = data_str.strip().split('\n')

# Process header
header = lines[0].split()

# Fix the "total plastic/ml" column name which is split
across multiple fields
if "total" in header and "plastic/ml" in header:
    total_index = header.index("total")
    header[total_index] = "total plastic/ml"
    header.remove("plastic/ml")

# Process data rows
for line in lines[1:]:
    parts = line.split()
    # The first two columns are p_r and ww_status
    processed_row = {
        'p_r': parts[0],
        'ww_status': parts[1]
    }

```

```

        # The rest are numeric columns: pH, turbidity,
conductivity, fibers, chunks, total
        processed_row['ph'] = float(parts[2])
        processed_row['turbidity'] = float(parts[3])
        processed_row['conductivity'] = float(parts[4])
        processed_row['avg_Fibers/ml'] = float(parts[5])
        processed_row['avg_chunks/ml'] = float(parts[6])
        processed_row['total plastic/ml'] = float(parts[7])

        rows.append(processed_row)

    # Create DataFrame from processed rows
    df = pd.DataFrame(rows)

    return df

#-----
# PART 1: DESCRIPTIVE STATISTICS & VISUALIZATION
#-----

def descriptive_statistics(df):
    """Generate descriptive statistics for the dataset."""
    print("==== DESCRIPTIVE STATISTICS ====")

    # Overall statistics
    print("\nOverall Statistics:")
    print(df.describe())

    # Group by p_r (Pool vs Riffle)
    print("\nStatistics by Location Type (Pool vs Riffle):")
    print(df.groupby('p_r')[['ph', 'turbidity', 'conductivity',

```

```

'avg_Fibers/ml',
                                'avg_chunks/ml', 'total
plastic/ml']]).describe())

# Group by ww_status (Above vs Below)
print("\nStatistics by Wastewater Status (Above vs Below):")
print(df.groupby('ww_status')[['ph', 'turbidity',
'conductivity', 'avg_Fibers/ml',
                                'avg_chunks/ml', 'total
plastic/ml']]).describe())

# Group by both p_r and ww_status
print("\nStatistics by Location Type and Wastewater
Status:")
print(df.groupby(['p_r', 'ww_status'])[['avg_Fibers/ml',
'avg_chunks/ml',
                                'total
plastic/ml']]).describe())

def correlation_analysis(df):
    """Analyze correlations between variables."""
    print("\n===== CORRELATION ANALYSIS =====")

    # Calculate correlation matrix
    corr_matrix = df[['ph', 'turbidity', 'conductivity',
'avg_Fibers/ml',
                                'avg_chunks/ml', 'total
plastic/ml']].corr()
    print("\nCorrelation Matrix:")
    print(corr_matrix)

    # Plot correlation heatmap

```

```

plt.figure(figsize=(10, 8))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', vmin=-
1, vmax=1)
plt.title('Correlation Matrix of Water Quality and
Microplastic Variables')
plt.tight_layout()
plt.savefig('correlation_heatmap.png')
plt.close()

print("\nCorrelation heatmap saved as
'correlation_heatmap.png'")

def visualize_data(df):
    """Create visualizations for the dataset."""
    print("\n==== DATA VISUALIZATION =====")

    # 1. Boxplots of microplastic content by location type
    plt.figure(figsize=(15, 5))

    plt.subplot(131)
    sns.boxplot(x='p_r', y='avg_Fibers/ml', data=df)
    plt.title('Fibers by Location Type')

    plt.subplot(132)
    sns.boxplot(x='p_r', y='avg_chunks/ml', data=df)
    plt.title('Chunks by Location Type')

    plt.subplot(133)
    sns.boxplot(x='p_r', y='total plastic/ml', data=df)
    plt.title('Total Plastic by Location Type')

    plt.tight_layout()

```

```

plt.savefig('microplastics_by_location.png')
plt.close()

# 2. Boxplots of microplastic content by wastewater status
plt.figure(figsize=(15, 5))

plt.subplot(131)
sns.boxplot(x='ww_status', y='avg_Fibers/ml', data=df)
plt.title('Fibers by Wastewater Status')

plt.subplot(132)
sns.boxplot(x='ww_status', y='avg_chunks/ml', data=df)
plt.title('Chunks by Wastewater Status')

plt.subplot(133)
sns.boxplot(x='ww_status', y='total plastic/ml', data=df)
plt.title('Total Plastic by Wastewater Status')

plt.tight_layout()
plt.savefig('microplastics_by_wastewater.png')
plt.close()

# 3. Interaction plots (location type x wastewater status)
plt.figure(figsize=(15, 5))

plt.subplot(131)
sns.boxplot(x='p_r', y='avg_Fibers/ml', hue='ww_status',
data=df)
plt.title('Fibers by Location and Wastewater Status')

plt.subplot(132)
sns.boxplot(x='p_r', y='avg_chunks/ml', hue='ww_status',

```

```

data=df)

plt.title('Chunks by Location and Wastewater Status')

plt.subplot(133)
sns.boxplot(x='p_r', y='total plastic/ml', hue='ww_status',
data=df)

plt.title('Total Plastic by Location and Wastewater Status')

plt.tight_layout()
plt.savefig('microplastics_interaction.png')
plt.close()

# 4. Scatter plots for water quality vs microplastic content
water_quality = ['ph', 'turbidity', 'conductivity']
microplastic = ['total plastic/ml']

for wq in water_quality:
    for mp in microplastic:
        plt.figure(figsize=(8, 6))
        sns.scatterplot(x=wq, y=mp, hue='ww_status',
style='p_r', data=df, s=100)
        plt.title(f'{mp} vs {wq}')
        plt.tight_layout()
        plt.savefig(f'{mp.replace("/", "_")}_vs_{wq}.png')
        plt.close()

print("\nVisualization plots saved as PNG files.")

#-----
# PART 2: STATISTICAL HYPOTHESIS TESTING
#-----

```

```

def hypothesis_testing(df):
    """Perform statistical hypothesis tests."""
    print("\n==== STATISTICAL HYPOTHESIS TESTING =====")

    # Define microplastic variables to test
    mp_vars = ['avg_Fibers/ml', 'avg_chunks/ml', 'total
plastic/ml']

    # 1. T-tests for comparison of means between Pool vs Riffle
    print("\nT-tests for Pool vs Riffle:")
    for var in mp_vars:
        pool_data = df[df['p_r'] == 'Pool'][var]
        riffle_data = df[df['p_r'] == 'Riffle'][var]

        # Check for normality (Shapiro-Wilk test)
        _, p_pool = stats.shapiro(pool_data)
        _, p_riffle = stats.shapiro(riffle_data)

        print(f"\n{var}:")
        print(f"    Shapiro-Wilk test for normality (Pool): p =
{p_pool:.4f}")
        print(f"    Shapiro-Wilk test for normality (Riffle): p =
{p_riffle:.4f}")

        # Perform t-test or non-parametric alternative
        if p_pool > 0.05 and p_riffle > 0.05:
            # Normal distribution, use t-test
            t_stat, p_val = stats.ttest_ind(pool_data,
riffle_data, equal_var=False)
            print(f"    T-test (assuming unequal variances): t =
{t_stat:.4f}, p = {p_val:.4f}")
        else:

```

```

        # Non-normal distribution, use Mann-Whitney U test
        u_stat, p_val = stats.mannwhitneyu(pool_data,
riffle_data)
        print(f"    Mann-Whitney U test: U = {u_stat:.4f}, p =
        {p_val:.4f}")

        print(f"    Mean Pool: {pool_data.mean():.4f}, Mean
Riffle: {riffle_data.mean():.4f}")

    # 2. T-tests for comparison of means between Above vs Below
wastewater
    print("\nT-tests for Above vs Below wastewater:")
    for var in mp_vars:
        above_data = df[df['ww_status'] == 'Above'][var]
        below_data = df[df['ww_status'] == 'Below'][var]

        # Check for normality
        _, p_above = stats.shapiro(above_data)
        _, p_below = stats.shapiro(below_data)

        print(f"\n{var}:")
        print(f"    Shapiro-Wilk test for normality (Above): p =
        {p_above:.4f}")
        print(f"    Shapiro-Wilk test for normality (Below): p =
        {p_below:.4f}")

        # Perform t-test or non-parametric alternative
        if p_above > 0.05 and p_below > 0.05:
            # Normal distribution, use t-test
            t_stat, p_val = stats.ttest_ind(above_data,
below_data, equal_var=False)
            print(f"    T-test (assuming unequal variances): t =

```

```

{t_stat:.4f}, p = {p_val:.4f}")
    else:
        # Non-normal distribution, use Mann-Whitney U test
        u_stat, p_val = stats.mannwhitneyu(above_data,
below_data)
        print(f" Mann-Whitney U test: U = {u_stat:.4f}, p =
{p_val:.4f}")

        print(f" Mean Above: {above_data.mean():.4f}, Mean
Below: {below_data.mean():.4f}")

    # 3. Two-way ANOVA to test interaction effects
    print("\nTwo-way ANOVA (Location Type x Wastewater
Status):")
    for var in mp_vars:
        # Make sure column names don't have special characters
for the formula
        df_copy = df.copy()
        if var == 'total plastic/ml':
            df_copy = df_copy.rename(columns={'total
plastic/ml': 'total_plastic_ml'})
            formula = f"total_plastic_ml ~ C(p_r) + C(ww_status)
+ C(p_r):C(ww_status)"
        elif '/' in var:
            new_var = var.replace('/', '_')
            df_copy = df_copy.rename(columns={var: new_var})
            formula = f"{new_var} ~ C(p_r) + C(ww_status) +
C(p_r):C(ww_status)"
        else:
            formula = f"{var} ~ C(p_r) + C(ww_status) +
C(p_r):C(ww_status)"

```

```

    model = ols(formula, data=df_copy).fit()
    anova_table = sm.stats.anova_lm(model, typ=2)

    print(f"\n{var}:")
    print(anova_table)

def run_complete_analysis():
    """Run the complete analysis pipeline."""
    # Load data
    df = load_data()
    print("Data loaded successfully. Shape:", df.shape)
    print(df.head())

    # Run descriptive statistics
    descriptive_statistics(df)

    # Run correlation analysis
    correlation_analysis(df)

    # Create visualizations
    visualize_data(df)

    # Run hypothesis tests
    hypothesis_testing(df)

    print("\nAnalysis complete!")

# Run the complete analysis
if __name__ == "__main__":
    run_complete_analysis()

```

Appendix B: Descriptive Statistics

Table 1. Summary Statistics for Water Quality Parameters and Microplastic Concentrations

Statistic	pH	Turbidity (NTU)	Conductivity (μ S/cm)	Fibers/ml	Chunks/ml	Total Plastic/ml
Mean	6.98	3.28	6.23	0.34	0.10	0.55
Std. Dev.	0.68	1.55	2.59	0.49	0.13	0.70
Min	5.58	1.61	3.77	0.03	0.00	0.04
25%	6.85	2.28	4.04	0.06	0.02	0.14
Median	7.26	2.71	5.65	0.13	0.04	0.24
75%	7.42	4.14	7.72	0.36	0.15	0.49
Max	7.64	7.33	12.34	1.65	0.46	2.56

Note: This table presents descriptive statistics for all measured variables across all 22 sampling sites

Table 2. Comparison of Water Quality Parameters and Microplastic Concentrations by Habitat Type

Variable	Pool (Mean $\pm$ SD)	Riffle (Mean $\pm$ SD)
pH	6.95 $\pm$ 0.74	7.00 $\pm$ 0.66
Turbidity	3.37 $\pm$ 1.72	3.19 $\pm$ 1.42
Conductivity	6.35 $\pm$ 2.73	6.11 $\pm$ 2.55
Fibers/ml	0.31 $\pm$ 0.50	0.38 $\pm$ 0.51
Chunks/ml	0.11 $\pm$ 0.14	0.09 $\pm$ 0.12
Total Plastic/ml	0.56 $\pm$ 0.80	0.54 $\pm$ 0.62

Note: This table compares mean values and standard deviations between pool (n=11) and riffle (n=11) habitats

Table 3. Comparison of Water Quality Parameters and Microplastic Concentrations by Wastewater Treatment Plant Position

Variable	Above WWTP (Mean ± SD)	Below WWTP (Mean ± SD)
pH	7.38 ± 0.21	6.64 ± 0.76
Turbidity	2.75 ± 0.99	3.72 ± 1.77
Conductivity	7.46 ± 3.25	5.20 ± 1.26
Fibers/ml	0.60 ± 0.65	0.13 ± 0.09
Chunks/ml	0.14 ± 0.17	0.06 ± 0.06
Total Plastic/ml	1.00 ± 0.85	0.18 ± 0.10

Note: This table compares mean values and standard deviations between sampling sites located upstream (n=10) and downstream (n=12) of the Ellensburg Wastewater Treatment Plant.

Table 4. Interaction Effects of Habitat Type and Wastewater Treatment Plant Position on Microplastic Concentrations

Location	WWTP Position	Total Plastic (Mean ± SD)	Fibers (Mean ± SD)	Chunks (Mean ± SD)
Pool	Above	1.03 ± 1.04	0.54 ± 0.65	0.14 ± 0.19
Pool	Below	0.17 ± 0.11	0.11 ± 0.08	0.07 ± 0.08
Riffle	Above	0.96 ± 0.73	0.67 ± 0.71	0.13 ± 0.17
Riffle	Below	0.18 ± 0.10	0.14 ± 0.10	0.05 ± 0.05

Note: This table presents the combined effects of habitat type and wastewater treatment plant position on microplastic metrics across all sampling sites.

Table 5. Correlation Matrix for Water Quality Parameters and Microplastic Metrics

Variable	pH	Turbidity	Conductivity	Fibers/ml	Chunks/ml	Total Plastic/ml
pH	1.00	0.08	0.14	0.24	0.42	0.37
Turbidity	0.08	1.00	-0.33	-0.01	-0.42	-0.27
Conductivity	0.14	-0.33	1.00	0.38	0.32	0.58
Fibers/ml	0.24	-0.01	0.38	1.00	-0.19	0.68
Chunks/ml	0.42	-0.42	0.32	-0.19	1.00	0.52
Total Plastic/ml	0.37	-0.27	0.58	0.68	0.52	1.00

Note: Values represent Pearson correlation coefficients (r) between all measured variables. Stronger correlations ($|r| > 0.5$) are indicated in bold.

Appendix C: Statistical Analysis of Factors Affecting Microplastic Distribution

Table 6. Mann-Whitney U Test Results Comparing Microplastic Concentrations Between Pool and Riffle Habitats

Microplastic Metric	U-statistic	p-value	Mean (Pool)	Mean (Riffle)
Fibers/ml	59.0	0.95	0.31	0.38
Chunks/ml	66.5	0.72	0.11	0.09
Total Plastic/ml	61.0	1.00	0.56	0.54

Note: Non-parametric tests were used due to non-normal distributions as determined by Shapiro-Wilk tests ($p < 0.05$).

Table 7. Statistical Comparison of Microplastic Concentrations Between Upstream and Downstream Locations

Microplastic Metric	Test Type	Statistic	p-value	Mean (Above)	Mean (Below)
Fibers/ml	Mann-Whitney U	84.0	0.12	0.60	0.13
Chunks/ml	Mann-Whitney U	66.5	0.69	0.14	0.06
Total Plastic/ml	T-test	3.03	0.01*	1.00	0.18

*Significant at $p < 0.05$

Note: Test type selected based on normality testing. Significant differences ($p < 0.05$) are indicated with an asterisk (*).

Appendix D: Two-way ANOVA Results for Microplastic Distributions

Table 8. Two-way ANOVA Results for Microplastic Fiber Concentrations

Factor	Sum of Squares	df	F-value	p-value
Location Type (p_r)	0.026	1	0.124	0.729
Wastewater Status	1.246	1	5.942	0.025*
Interaction	0.015	1	0.071	0.792
Residual	3.774	18	-	-

* $p < 0.05$, ** $p < 0.01$

Note: Analysis of variance testing the effects of habitat type, wastewater treatment plant position, and their interaction on fiber concentrations. Significant effects ($p < 0.05$) are indicated with an asterisk (*).

Table 9. Two-way ANOVA Results for Microplastic Fragment (Chunk) Concentrations

Factor	Sum of Squares	df	F-value	p-value
Location Type (p_r)	0.002	1	0.124	0.729
Wastewater Status	0.032	1	1.868	0.189
Interaction	0.000	1	0.026	0.875
Residual	0.306	18	-	-

* $p < 0.05$, ** $p < 0.01$

Note: Analysis of variance testing the effects of habitat type, wastewater treatment plant position, and their interaction on fragment concentrations.

Table 10. Two-way ANOVA Results for Total Microplastic Concentrations

Factor	Sum of Squares	df	F-value	p-value
Location Type (p_r)	0.003	1	0.008	0.928
Wastewater Status	3.658	1	9.949	0.005**
Interaction	0.009	1	0.024	0.879
Residual	6.617	18	-	-

* $p < 0.05$, ** $p < 0.01$

Note: Analysis of variance testing the effects of habitat type, wastewater treatment plant position, and their interaction on total microplastic concentrations. Significant effects ($p < 0.01$) are indicated with a double asterisk (**).